B- Development Methodology – Scrum

By Stack Architects

1. Sprint 2 Overview

Sprint Duration: 25 Days (with eid holidays)

Sprint Goal:
The goal of Sprint 2 was to implement the core operational functionality of the Library Management System. This included developing a complete book catalogue system, enabling students to search and request books, implementing a controlled borrowing workflow with librarian approval, and assigning due dates to issued books. The sprint also ensured proper tracking of individual book copies and enforcement of borrowing constraints.

Modules Developed in Sprint 2:

- Book Catalogue Management Module
- Book Search and Viewing Module
- Borrow Request Management Module
- Borrow Approval and Control Module
- Due Date and Borrow Tracking Module

2. Sprint 2 Backlog

a) User Stories

US-01 – Add New Books to Catalog
As a librarian, I want to add new books so that they become available in the system.

US-02 – Update Book Details
As a librarian, I want to update book details so that information remains accurate.

US-03 – Delete Book
As a librarian, I want to delete books so that outdated records are removed.

US-04 – Book Search
As a student, I want to search books so that I can find relevant titles easily.

US-05 – Submit Borrow Request
As a student, I want to request a book so that I can borrow it.

US-06 – View Pending Borrow Requests
As a librarian, I want to view pending requests so that I can process them.

US-07 – Approve Borrow Request
As a librarian, I want to approve a borrow request so that a book is issued and due date is assigned.

US-08 – Reject Borrow Request
As a librarian, I want to reject invalid requests so that incorrect borrow attempts are prevented.

US-09 – View Borrow History
As a student, I want to view my borrow history so that I can track issued books and due dates.

US-10 – Borrow Limit Enforcement
As a system, I want to enforce a borrowing limit so that resources are fairly distributed.

US-11 – Unique Book Copy Tracking
As a librarian, I want each book copy to have a unique identifier so that individual copies can be tracked.

b) Sub-User Stories and Task Breakdown

US-01 – Add New Books

- Design books table schema
- Design book copies table
- Create Book entity
- Create BookCopy entity
- Implement BookService logic
- Create REST API for adding books
- Create UI form for librarian
- Generate unique ISBN and copy codes

US-02 – Update Book Details

- Implement update logic
- Create update API

- Create UI edit form
- Validate changes

US-03 – Delete Book

- Implement delete logic with validation
- Ensure all copies are available before deletion
- Create delete API
- Create UI confirmation

US-04 – Book Search

- Create search queries
- Implement search API
- Design search UI
- Display filtered results

US-05 – Submit Borrow Request

- Design borrow_requests table
- Create BorrowRequest entity
- Implement request API
- Add validation checks
- Prevent duplicate requests
- Create student request UI

US-07 – Approve Borrow Request

- Design borrow_records table
- Create BorrowRecord entity
- Implement approval logic
- Reduce available book count
- Assign book copy
- Calculate due date
- Create approval API

US-10 – Borrow Limit Enforcement

- Add borrow limit check (max 3 books)
- Implement validation logic
- Display error messages

3. Structured Specifications

1. Add New Book (US-01)

Inputs:
Book title, author, category, total copies

Outputs:
New book created with unique ID and copy records

Processing:
System validates input, generates unique book ID and ISBN, creates multiple book copies, and stores data in the database

Preconditions:
Librarian must be logged in

Postconditions:
Book and its copies are stored and available in the system

Exceptions:
Invalid input, duplicate data, validation failure

2. Submit Borrow Request (US-05)

Inputs:
Student ID, Book ID, requested duration

Outputs:
Borrow request created with PENDING status

Processing:
System validates request, checks availability, prevents duplicates, and stores request in database

Preconditions:
Student must be logged in

Postconditions:
Borrow request is stored and visible to librarian

Exceptions:
Book unavailable, duplicate request, borrowing limit exceeded

3. Approve Borrow Request (US-07)

Inputs:
Borrow request ID

Outputs:
Borrow request approved and record created

Processing:
System updates request status, assigns a book copy, decreases available quantity, and calculates due date

Preconditions:
Librarian must be logged in and request must be pending

Postconditions:
Borrow record created with due date

Exceptions:
No available copies, invalid request, concurrency issues

4. Book Search (US-04)

Inputs:
Search parameters (title, author, category)

Outputs:
Filtered list of books

Processing:
System executes search queries on database and returns matching results

Preconditions:
User must be logged in

Postconditions:
Search results displayed to user

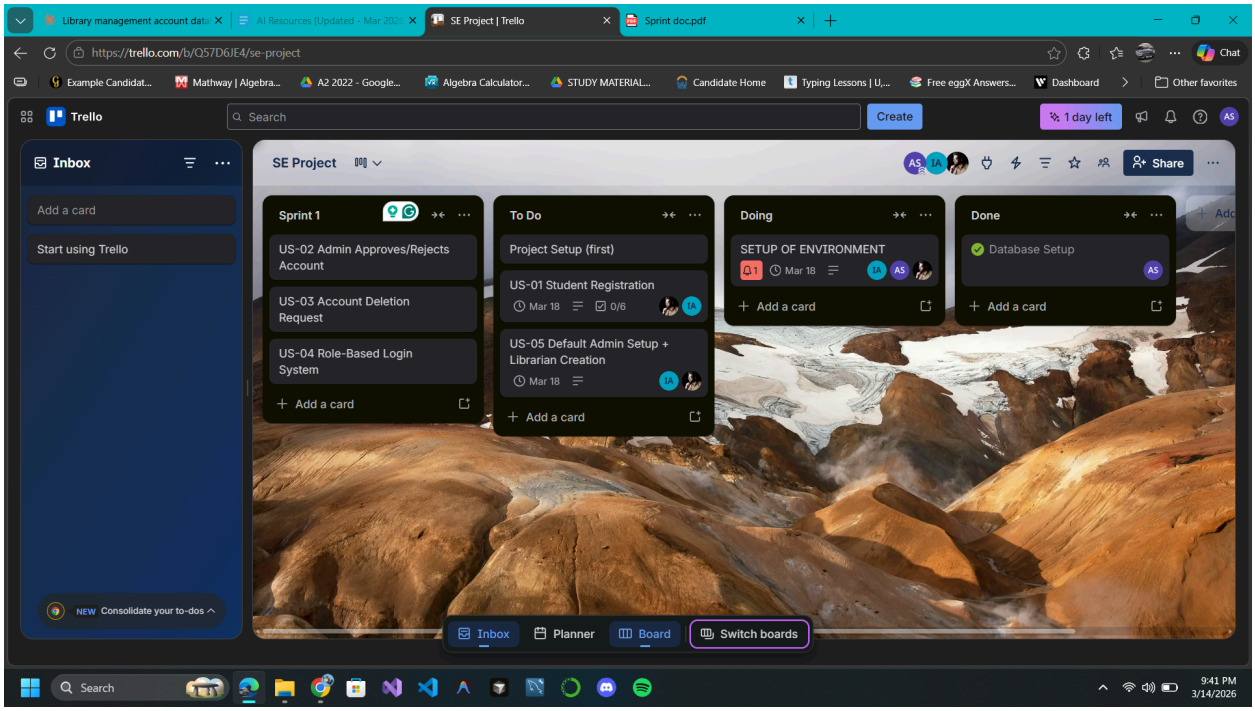
Exceptions:
No results found, invalid query

4. Scrum Board

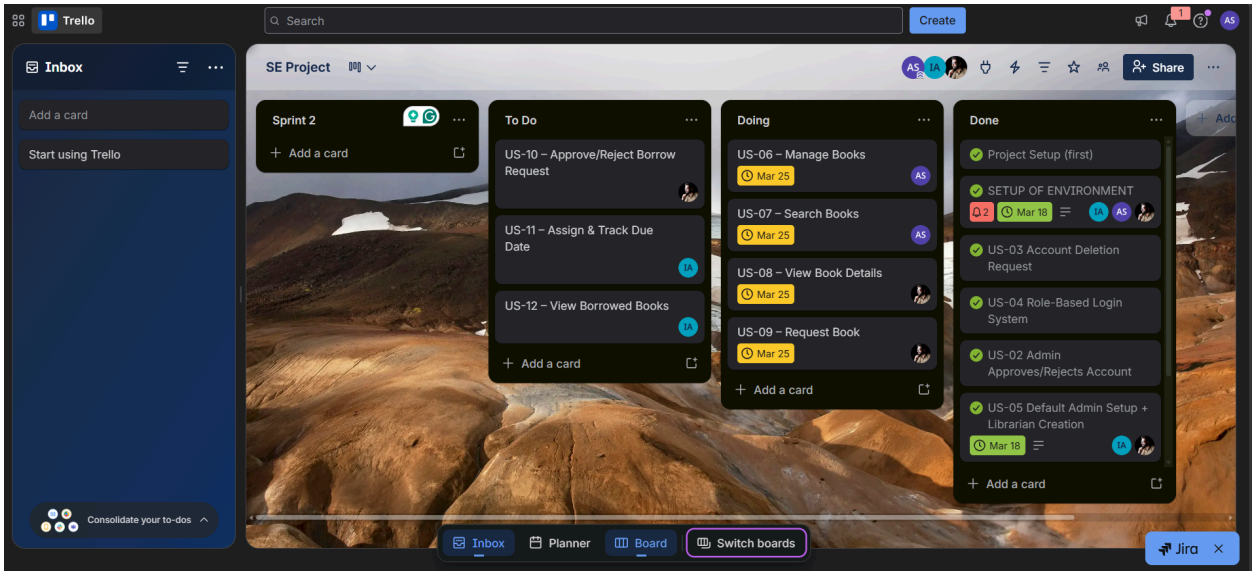
The Scrum board was maintained using Trello to track sprint progress across different stages.

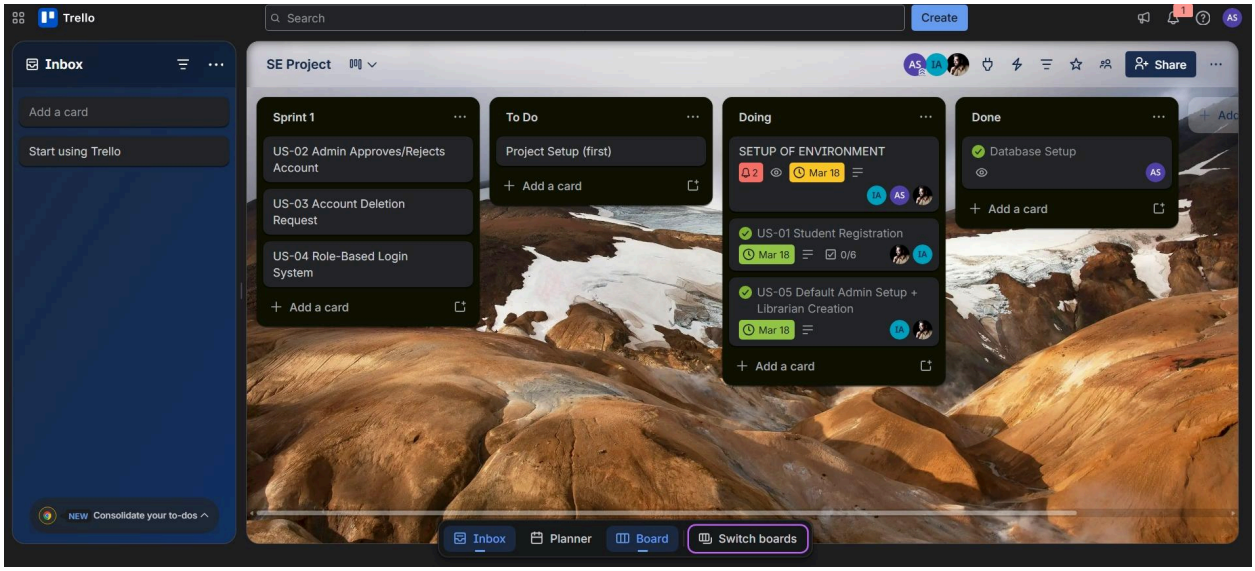
Trello Board Snapshots:

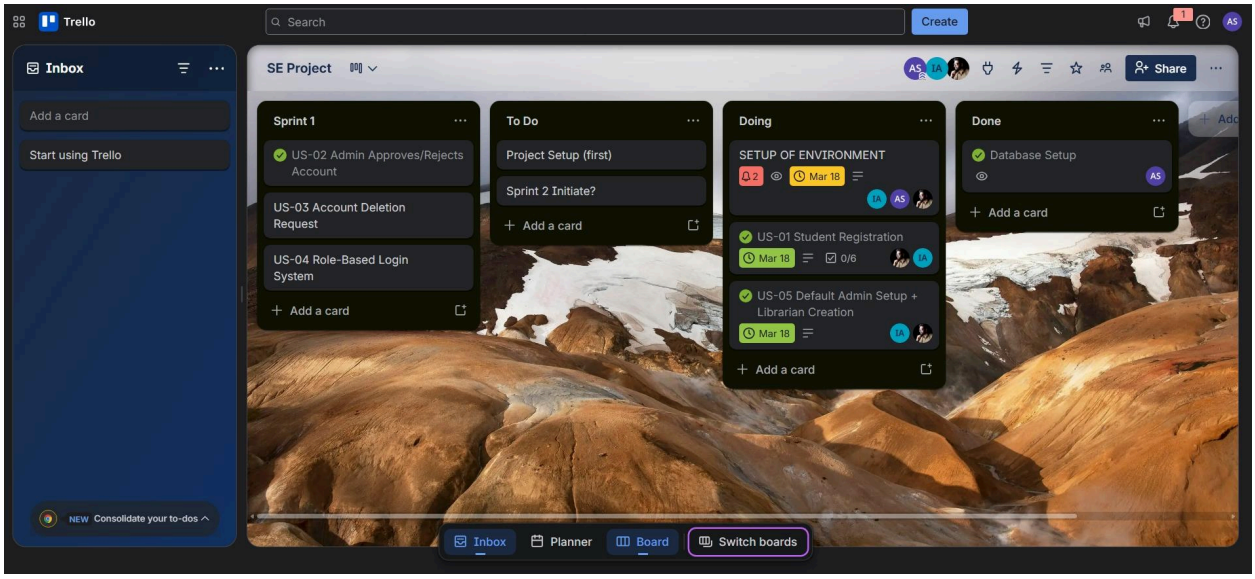
- Sprint Planning Stage:



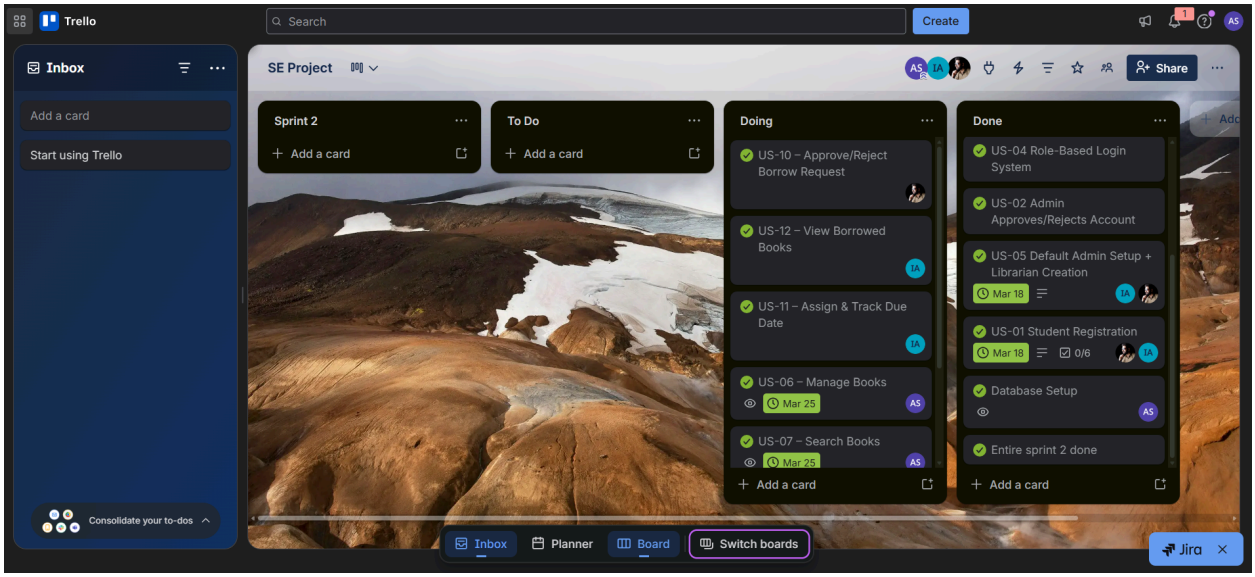
- Mid-Sprint Progress:







- Sprint Completion Stage:



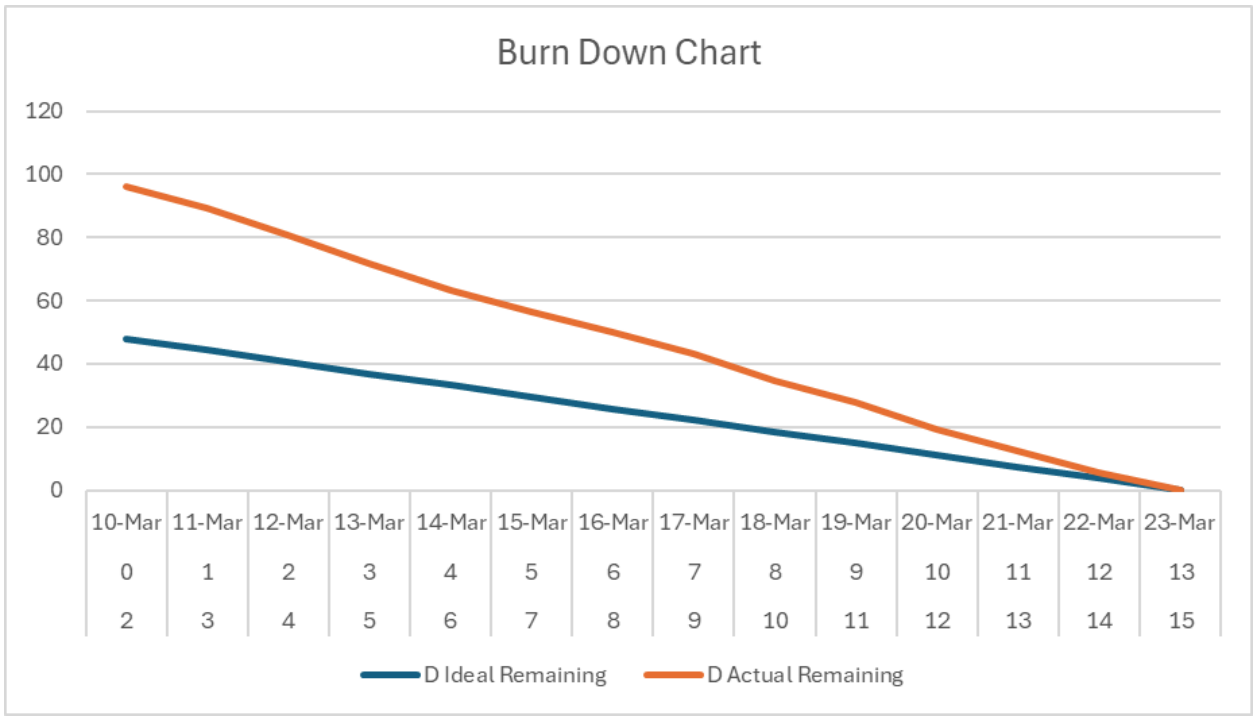
The board was organized into the following columns:

- Backlog
- To Do
- In Progress
- Done

All user stories were tracked and updated regularly throughout the sprint

5. Sprint 2 Burn-Down Chart

A burn-down chart was used to track the progress of Sprint 2 by comparing planned work versus completed work over time.



The chart illustrates:

- Total planned story points: 48
- Completed story points: 48
- Completion rate: 100%

All planned user stories were successfully completed within the sprint duration, with no leftover tasks.